

ANTÓNIO B. LINHARES OLIVEIRA

antonioliveira@protonmail.com · (+351) 968 615 284

Porto, Portugal

github.com/antonoliv

PROFILE

A detail-oriented and extroverted graduate fascinated by software development, data science and solving complex problems. An unconventional and creative thinker with excellent communication and conflict resolution skills acquired through years of extensive group work during college, where I often took a leadership role. Comfortable in an international and interdisciplinary environment and over a year of experience as an association manager.

EXPERIENCE

Euronext Technologies

October 2024 - Present

AI Engineer

Porto

- Designed and deployed an internal, language-agnostic Documentation Generation application targeting legacy codebases (COBOL, EGL, PL/1), significantly accelerating documentation efforts for teams facing extensive backlogs of undocumented legacy code.
- Core maintainer of LLM Gateway, a Java microservice acting as the abstraction layer between internal systems and AI providers (Azure OpenAI, Amazon Bedrock), ensuring reliability, extensibility, and provider-agnostic integration.
- Core maintainer of RAG Platform, a Python microservice leveraging LlamaIndex to expose RAG indexing and querying capabilities to internal applications, enabling semantic search and retrieval-augmented generation at scale.
- Contributed to the development of MCP Gateway, a proxy application bridging remote and locally deployed MCP servers with developer tools such as GitHub Copilot — incorporating centralized SSO authentication, guardrails, and usage monitoring to abstract deployment and integration complexity from end users.
- Drove key engineering standards across the team: pioneered the first fully production-deployed project using GitLab CI/CD and Terraform, and championed UV adoption for Python dependency management — reducing deployment overhead and freeing team capacity for development and innovation.
- **Technologies:** Python · Docker · AWS · Azure · Terraform · Gitlab CI/CD · Amazon Bedrock · Azure OpenAI API · MCP

Natixis

April 2022 - July 2022

Software Engineer Intern

Porto

- Curricular Internship for the Bachelor's degree in Informatics Engineering of ISEP
- Development of a web application and data pipeline that integrates information from different platforms and processes of the company to facilitate the operations of a support team in the company
- **Technologies:** PHP · jQuery · JavaScript · HTML/CSS · Bootstrap · SQL Server
- **Grade:** 16 / 20

EDUCATION

Faculdade de Engenharia da Universidade do Porto

September 2022 - September 2024

Master's in Informatics and Computing Engineering

- Dissertation (in English) on "Graph Reinforcement Learning for improving Smart Grid Services", focusing on combining Reinforcement Learning algorithms with Graph Neural Networks to enhance intelligent system's performance on graph-structured environments. Developed with Python, Gymnasium, PyTorch, Stable-Baselines3 and other libraries
- **Dissertation Grade:** 19 / 20
- **Overall Grade:** 17 / 20

Instituto Superior de Engenharia do Porto
Bachelor's in Informatics Engineering

September 2019 - June 2022

- European Project Semester (ERASMUS) at Saxion University of Applied Sciences, in Enschede, Netherlands
- Member of Tuna Académica do ISEP
- **Overall Grade:** 15 / 20

TECHNICAL STRENGTHS

Languages	Portuguese (Native) · English (C2) · Spanish (B1) · French (A1)
Programming	Python · Java · TypeScript · JavaScript · Kotlin · C · PHP · R · SQL · HTML/CSS · Bash
Generative AI	Prompt Engineering · Agentic Systems · RAG · MCP · LlamaIndex · Amazon Bedrock · Azure OpenAI · GitHub Copilot · Claude Code · Opencode
Machine Learning	PyTorch · scikit-learn · Gymnasium · pandas
Backend & APIs	FastAPI · SQLAlchemy · SpringBoot · JPA · REST · MVC
DevOps & Cloud	Linux · Docker · Terraform · GitLab CI/CD · AWS ECS · AWS Lambda
Databases	PostgreSQL · MySQL · Oracle Database · Microsoft SQL Server · MongoDB · AWS S3

PROJECTS

Laboratory of Project Management (LGP) February 2023 - June 2023
Smartwatch Triggered AI Enhanced (Sports) Video Highlight Experience FEUP

- Developed in partnership with MOG Technologies and the context of the LGP course of the Masters in Informatics and Computing Engineering of FEUP with a team of 11 students
- Idealization of a startup, H4PRO (Highlights for PRO), around the main requested product: A smart-phone and smartwatch application that connected with a visual sensor ecosystem in Padel fields and allowed the players to keep track of game score and save game highlights on-demand
- Assumed the role of the startup's CTO and also took charge of wearable development
- Sources: <https://github.com/LGP-FEUP-2023/SC12>
- Pitch: <https://www.youtube.com/watch?v=d5L8K53Xqys>
- **Technologies:** React Native · Kotlin
- **Grade:** 18 / 20 (Highest of the group)

Large-Scale Software Development (DS) September 2022 - February 2023
Circular Economy for the Built Environment Platform FEUP

- Carried out in partnership with Built COLAB and the context of the DS course of the Masters in Informatics and Computing Engineering of FEUP with a team of 23 students sub-divided into groups of 5/6 members
- Developed a web platform for connecting sellers of raw materials produced from construction, renovation or demolition works with buyers that can bring new value to those same materials.
- **Technologies:** FastAPI · Python · Vue.js · Docker · SQL · MVC · REST
- **Grade:** 16 / 20

European Project Semester September 2021 - February 2022
Data-driven extended-life platform for reusable consumer electronics Saxion University

- Developed during ERASMUS mobility at Saxion University of Applied Sciences in Enschede, Netherlands with a team of 6 students from different academic backgrounds and nationalities
- Research project on how electronic appliances and components are collected and repurposed. Met with several local Dutch appliance stores to study how the process can be optimized and scaled with information systems

- Design and development of a proof-of-concept stock system composed of a REST API, web interface and a simple HTTP server
- Sources: <https://github.com/antonoliv/smartsolutions>
- **Technologies:** Java · HTML/CSS · JavaScript · Docker · SQL · MVC · REST
- **Grade:** 8 / 10 (Highest of the group)